BRAB PUTT – Bi-Terrain Rover for Autonomous Beer-Can Pursuit

Autonomous Target Tracking Rover for Outdoor Litter Collection

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Abstract

This poster presents the development of BRAB PUTT (Bi-Terrain Rover for Autonomous Beer-Can Pursuit and Utilising Target Tracking). A compact mobile robot designed to assist with outdoor litter collection. The system autonomously searches, detects, and pursues aluminium drink cans in challenging marsh-like environments.

The rover is built on a lightweight 200 mm × 200 mm chassis constructed from cast acrylic and PLA composites and equipped with large off-road wheels for enhanced terrain handling. A four-wheel-drive configuration powered by 12 V encoder motors enables stable movement and accurate feedback control through PID regulation. Vision-based object detection and state-driven navigation behaviours were integrated to allow the rover to locate a target can, traverse uneven ground, and return to its start point.

The project demonstrates how cybernetic principles, state-based control, and object detection can be combined in a small, low-cost platform intended for autonomous outdoor search tasks.

Introduction

More traditional mobile robots could rely on remote control, where the robot's movement decisions are given by a human operator. Whereas in the autonomous machine we've built, the systems must simultaneously perform both perception and decision-making internally. This required a great integration of sensing, interpretation, planning, and action.

BRAB PUTT was developed to explore this level of autonomy without indoor comforts (dry ground, dry atmosphere, constant and clear lighting). Whereas with it going outside added a level of complexity as now the robot must: identify areas to explore with inconsistent and always changing environments, navigate through uneven, wet, or soft terrain, detect a unique object (aluminium coke cans) using computer vision, pursue and approach the detected object, and return to the origin.

The project draws on core concepts in Robotics and Cybernetics: feedback loops for real-time control, state estimation for tracking the robot's condition, behavioural states for structuring navigation logic, and visual recognition for object detection. The aim is to demonstrate how these frameworks can be showcased in a small, rapidly prototyped machine platform suited to unpredictable outdoor environments.

Main Objectives

The project was built around these objectives:

The Design and construction: a lightweight bi-terrain mobile robot capable of operating on obstacle-filled, marshy, uneven ground.

Implement Four Wheel Drive locomotion using encoders for PID control.

Develop computer vision for object detection. It must be able to detect aluminium drink cans with inconsistent lighting and conditions.

Create navigation algorithms and behavioural states for searching, tracking, pursuing, and returning.

Error handling in real outdoor conditions. Showcasing to the user any errors within or outside of itself.

Materials and Methods

The Structure

The robot chassis 200 mm × 200 mm is constructed using cast acrylic and PLA to minimise weight, cost, and keep a solid structure. They're: lightness helped reduce motor load. The ease of 3d printing and laser cutting assisted rapid fabrication and modification.

It has moderate environmental durability is suitable but the main point of using it was for prototyping and it's cost effective nature. Large RC off-road wheels with large surface area were chosen for greater contact area and improved traction on grass or soft ground.

Locomotion System

A four-wheel-drive was implemented using 4 12 V worm-gear encoder motors for: high torque for overcoming soft terrain, integrated encoders enabling precise PID-based speed control, and a simple mechanical layout with few moving parts.

Power was supplied using a 12 V Nickel battery, providing a well-tested and compact solution for continuous outdoor operation and to fit into a small machine.

Motor control was achieved using two L298N H-bridge drivers, with the motor voltage and current requirements that they provide, and because I had used them before, I already knew how to utilise them.

Electronics and Software

A microcontroller as the brain that would implement: PID motor control loops, state-machine navigation logic, encoder feedback processing, and serial communication with the vision module.

Computer Vision

Using a Pixy2 or Raspberry Pi camera module 3, the object detection was implemented using feature-based recognition set to red for Coke cans, then built up from there. Outdoor specific considerations such as glare, shadows, and varying lighting. We addressed this by purchasing a higher-quality and premium component that would already be applying thresholding, shape filtering, and machine learning.

Navigation and States

The rover operated under the state machine architecture inspired by Brooks' subsumption model. The States implemented:

- Search: move through the environment following a sweep pattern.
- Detect: verify visual identification of a drink can.
- Pursue: align heading and drive toward the target.
- Acquire: stop at close range for retrieval.
- Return: retrace the stored path using reversed movement logs.

This approach reduced computational complexity by keeping the states to a minimum, helping to reduce the workload by decreasing the amount we would have to program and test.

Mathematical Section

PID Control

Motor speed was regulated using the classical PID control equation: The error between the target and measured motor speed, proportional, integral, and derivative gains. PID tuning was performed empirically to achieve minimal overshoot and reliable steady-state operation in outdoor conditions.

Path Reversal

Wheel encoders enabled step-wise estimation of distance travelled. The path was stored as a sequence of movement commands and reversed during the return phase.

$$E = mc^2 \tag{1}$$

Results

Testing initially indoors, connected to a power supply later, was completed in controlled outdoor conditions featuring wet and uneven terrain. Key observations included:

Overall, BRAB PUTT demonstrated reliable, simultaneous, autonomous operation. Validating the integration of cybernetic principles in a compact field-testable rover.

Conclusions

This project highlighted the effectiveness of an iterative, bottom up approach to autonomous robot development. By first establishing robust locomotion, then feedback control, and then with state-machine logic. Integrating machine vision and navigation later reduced the risk of compounding lengthy design errors.

The resulting rover achieved the primary goal of autonomously detecting and approaching drinks cans in outdoor environments. The work also emphasised the importance of quick and cheap durable materials, precise motor control, and structured behavioural design for off-road autonomous systems.

Forthcoming Research

Future work will focus on increasing robustness and expanding capabilities:

Rubber sealed enclosures for improved water protection in harsher wetter weather.

More variable object detection to detect more than just drinks can.

Alternative drive configuration for steering. Adding steering columns so that this design could work with a variety of motors.

References

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